

exercice_en

October 8, 2026

```
[3]: import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

data = [
    14.0, 7.6, 11.2, 12.8, 12.5, 9.9, 14.9, 9.4, 16.9, 10.2, 14.9, 18.1, 7.3, 9.
↪8, 10.9,
    12.2, 9.9, 2.9, 2.8, 15.4, 15.7, 9.7, 13.1, 13.2, 12.3, 11.7, 16.0, 12.4, ↪
↪17.9, 12.2,
    16.2, 18.7, 8.9, 11.9, 12.1, 14.6, 12.1, 4.7, 3.9, 16.9, 16.8, 11.3, 14.4, ↪
↪15.7, 14.0,
    13.6, 18.0, 13.6, 19.9, 13.7, 17.0, 20.5, 9.9, 12.5, 13.2, 16.1, 13.5, 6.3, ↪
↪6.4, 17.6,
    19.1, 12.8, 15.5, 16.3, 15.2, 14.6, 19.1, 14.4, 21.4, 15.1, 19.6, 21.7, 11.
↪3, 15.0,
    14.3, 16.8, 14.0, 6.8, 8.2, 19.9, 20.4, 14.6, 16.4, 18.7, 16.8, 15.8, 20.4, ↪
↪15.8, 22.4,
    16.2, 20.3, 23.4, 12.1, 15.5, 15.4, 18.4, 15.7, 10.2, 8.9, 21.0
]

s = pd.Series(data)

print(f"Minimum : {s.min()}")
print(f"Écart-type : {s.std():.2f}")
print(f"Médiane : {s.median():.2f}")
print(f"Maximum : {s.max()}")
print(f"Moyenne : {s.mean():.2f}")

fig, (ax1, ax2) = plt.subplots(2, 1, figsize=(8, 10))

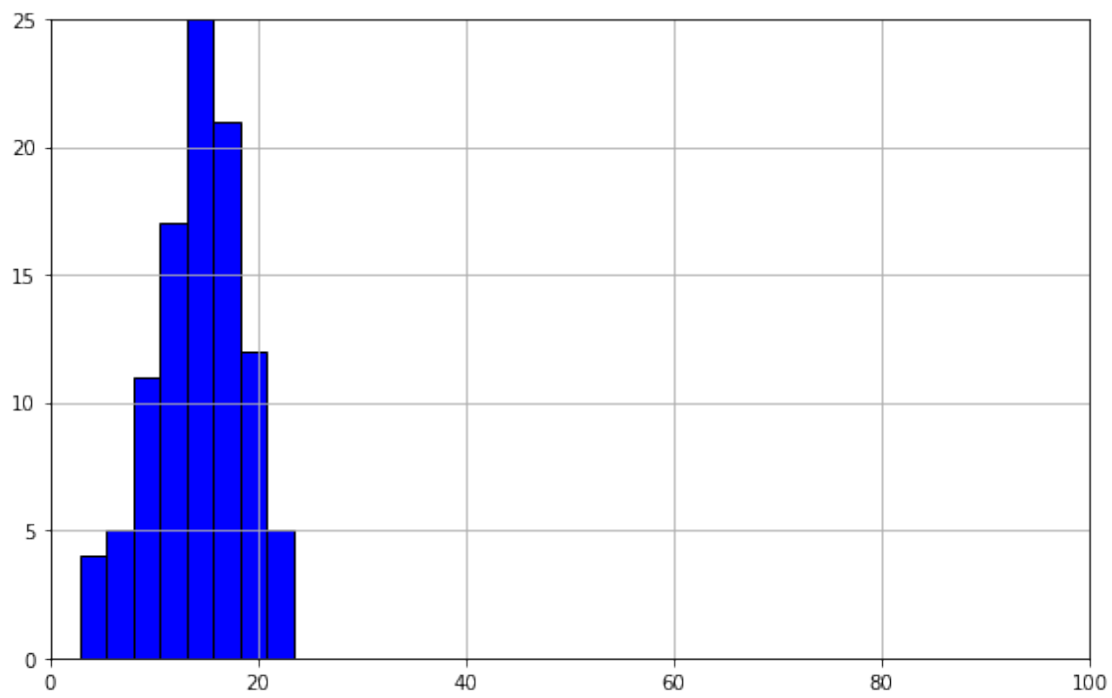
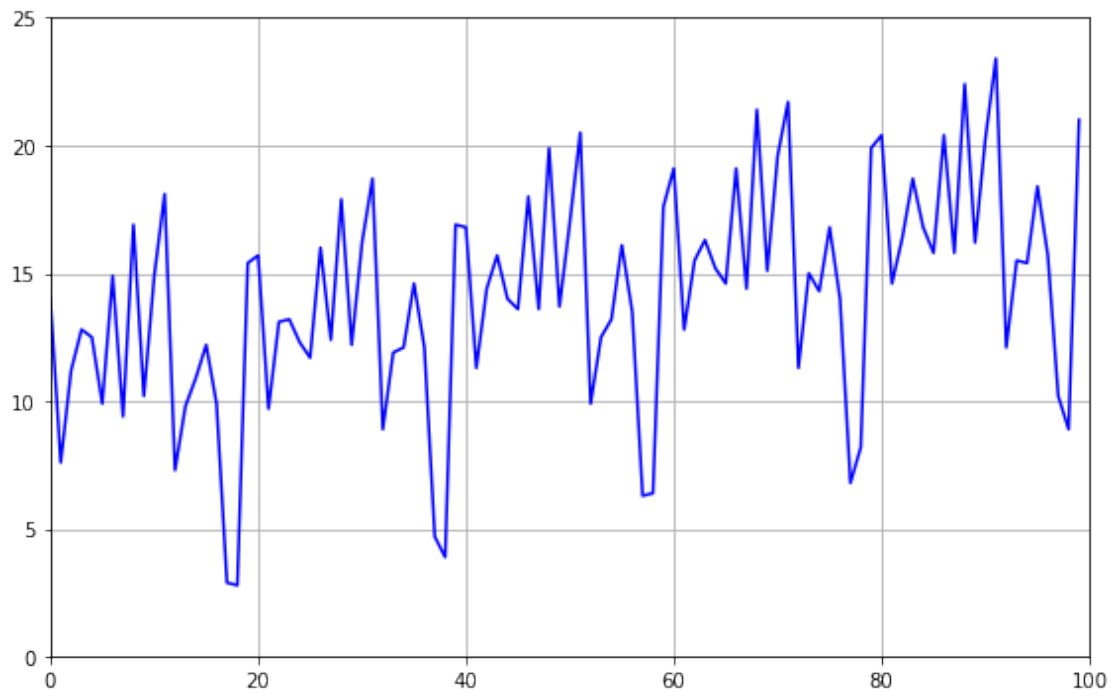
ax1.plot(data, color='blue')
ax1.set_xlim(0, 100)
ax1.set_ylim(0, 25)
ax1.grid(True)

ax2.hist(data, bins=8, color='blue', edgecolor='black')
ax2.set_xlim(0, 100)
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ax2.set_ylim(0, 25)
ax2.grid(True)

plt.tight_layout()
plt.show()
```

Minimum : 2.8
Écart-type : 4.33
Médiane : 14.50
Maximum : 23.4
Moyenne : 14.11



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